

Mathematics - Similar Triangles, Coordinate

Geometry...

1. $\triangle ABC$ and $\triangle DCB$ are similar

$$\therefore \frac{AB}{AC} = \frac{DC}{DB} \Rightarrow AB \times DB = DC \times AC$$

2. Let (a, b) be any point in the plane.

Distance from origin of (a, b) is $(0, 0)$

$$\text{Distance Formula} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(a - 0)^2 + (b - 0)^2}$$

$$= \sqrt{a^2 + b^2}$$

$$= \sqrt{a^2 + b^2}$$

3. In $\triangle ABC$ and $\triangle DAC$

$$\angle BAC = \angle ADC \text{ — (Given)}$$

$$\angle ACB = \angle ACD \text{ — (common)}$$

$\therefore \triangle ABC \sim \triangle DAC$ — by AA similarity

$$\Rightarrow \frac{AC}{BC} = \frac{DC}{AC}$$

$$4. \quad x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

$$5. \quad \text{To find} - \frac{\text{ar}(\triangle DEF)}{\text{ar}(\triangle ABC)}$$

$$\therefore FD = \frac{1}{2} BC$$

$$ED = \frac{1}{2} AC$$

$$EF = \frac{1}{2} AB$$

In $\triangle ABC$ and $\triangle EFD$,

$$\frac{AB}{EF} = \frac{BC}{FD} = \frac{AC}{ED} = 2 \quad \text{--- (1)}$$

$\triangle ABC \sim \triangle EFD$ -- (SSS similarity)

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle EFD)} = \left(\frac{AB}{EF}\right)^2 = 4$$

$$\frac{+mz y_1}{+mz}$$

[from 9]

$$\frac{\text{ar}(\triangle EFD)}{\text{ar}(\triangle ABC)} = \frac{1}{4}$$

∴ The area is 1:4.

6. Given - A (-3, 2), B (6, 1)
 $x_1 \ y_1 \quad x_2 \ y_2$

let the coordinate of points (0, y) divide the ratio m:n

$$(0, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{6m - 3n}{m+n}, \frac{m + 2n}{m+n} \right) = (0, y)$$

$$\therefore \frac{6m - 3n}{m+n} = 0$$

$$\therefore 6m = 3n \quad \text{ie} = \frac{m}{n} = \frac{1}{2}$$

Ratio = 1:2

7. In $\triangle ADE$ & $\triangle ABC$

$$\angle A = \angle A \text{ — (Common)}$$

$$\angle EDA = \angle CBA \text{ (Corresponding angles)}$$

Hence, by AA similarity,

$\triangle ADE$ is similar to $\triangle ABC$

Similarity of Triangles -

$$\frac{AD}{AB} = \frac{DE}{BC} = \frac{AE}{AC}$$

$$\frac{AD}{AD+BD} = \frac{AE}{AE+EC}$$

$$\frac{AD+BD}{AD} = \frac{AE+EC}{AE}$$

$$1 + \frac{BD}{AD} = 1 + \frac{EC}{AE}$$

$$\frac{BD}{AD} = \frac{EC}{AE}$$

Substituting -

$$\frac{x+4}{2x+7} = \frac{2x-1}{3x}$$

$$\frac{BD}{AD} = \frac{EC}{AE}$$

$$\frac{x+3}{x+4} = \frac{x+1}{2x-1}$$

$$2x^2 + 5x - 3 = x^2 + 5x + 4$$

$$x^2 = 7$$

$$x = \sqrt{7}$$

g. $AD^2 = AB^2 + BD^2$ ($\triangle ABD$ is a $\angle 90^\circ$ at B)

$$AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad [\because BD = DC]$$

$$AD^2 = AB^2 + \frac{1}{4} BC^2 \quad \dots (i)$$

($\triangle BCE$ is $\angle 90^\circ$ at B)

$$\therefore CE^2 = BC^2 + BE^2$$

$$CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2 \quad [\because BE = EA]$$

$$CE^2 = BC^2 + \frac{1}{4} AB^2 \quad \dots (ii)$$

Add (i) and (ii)

$$AD^2 + CE^2 = AB^2 + \frac{1}{4} BC^2 + BC^2 + \frac{1}{4} AB^2$$

$$\Rightarrow AD^2 + CE^2 = \frac{5}{4} (AB^2 + BC^2)$$

$$AD^2 + CE^2 = \frac{5}{4} (AC^2) \quad [\because ABC \text{ is right triangle}]$$

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$$\therefore [AC^2 = AB^2 + BC^2]$$

$$\Rightarrow \left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$$

$$CE^2 = \frac{125}{4} - \frac{45}{4} = 20$$

$$CE = \sqrt{20} \text{ cm} = 2\sqrt{5} \text{ cm.}$$

10. Area of $\triangle ABC$

$$= \frac{1}{2} [2(4-6) + 4(6-2) + 2(2-4)]$$

$$= \frac{1}{2} [2(-2) + 4(4) + 2(-2)] = 4 \text{ sq. units.}$$